Journal of Power Sources 677 (2026) 239941

Contents lists available at ScienceDirect

Journal of Power Sources

journal homepage: www.elsevier.com/locate/jpowsour

A hybrid framework for PEMFC degradation prediction integrating Crossformer and physical constraints☆

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HIGHLIGHTS

- A hybrid Crossformer–PINN framework for PEMFC degradation prediction is proposed.
- Three-stage training strategy enables synergistic data-physics optimization.
- Physics-informed constraints enhance robustness under limited training data.

ARTICLE INFO

Keywords:
Proton exchange membrane fuel cell
Hybrid prediction model
Physics-informed neural network
Performance degradation prediction

ABSTRACT

Durability remains a critical barrier to the large-scale commercial adoption of proton exchange membrane fuel cells (PEMFCs). To address the limitations of pure data-driven methods in physical interpretability and model-driven approaches in prediction accuracy, this paper proposes a hybrid prediction framework integrating Crossformer with physics-informed learning for PEMFC degradation voltage forecasting. The framework employs convolutional layers and Crossformer's cross-dimensional attention mechanism to effectively extract local temporal features and model dependencies among multiple parameters, enabling accurate characterization of the dynamic voltage decay behavior. Moreover, by incorporating electrochemical mechanisms as physical constraints and adopting a three-stage progressive training strategy, it achieves synergistic optimization between data-driven learning and physical modeling. Experimental results on three aging datasets under steady-state, quasi-dynamic, and dynamic current conditions demonstrate that the proposed model achieves superior generalization performance across various operating scenarios. The model attains root mean square error (RMSE) values of 0.00191–0.00305 V and mean absolute percentage error (MAPE) values of 0.0339%–0.323%, outperforming Informer, convolutional neural network (CNN), and other comparative models. Ablation studies further confirm the individual contributions of the convolutional module and physical constraints to prediction accuracy, as well as the model's robustness under limited training data conditions.

1. Introduction

With the ongoing intensification of the global energy crisis and increasing demands for environmental protection, the development of efficient and clean new energy technologies has become a major focus of scientific research worldwide [1]. Proton exchange membrane fuel cells (PEMFCs) are widely regarded as a key component of future sustainable energy systems due to their high energy density, fast dynamic response, and zero emissions [2], demonstrating broad potential in electric vehicles, portable power sources, and distributed power generation [3]. However, performance degradation and limited lifespan remain major obstacles to their large-scale commercial application [4]. Accurate prediction of PEMFC health status is crucial for developing sound maintenance strategies, extending service life, reducing operational costs, and improving overall system reliability [5].

The performance of PEMFCs fundamentally depends on the electrochemical properties of their core materials. Recent advances in novel electrocatalysts, such as CuTe supported on graphitic carbon nitride nanocomposites for efficient water oxidation in alkaline media [6], have opened new avenues for improving electrode kinetics. Concurrently, functional materials with tailored structural characteristics, including hexaferrite compounds exhibiting stable performance at elevated temperatures [7] and perovskite-type oxides with optimized

☆ This work was supported in part by the National Natural Science Foundation of China under Grant 62503449, in part by Anhui Provincial Natural Science Foundation, China under Grant 2508085QF215, and in part by the Fundamental Research Funds for the Central Universities, China under Grant WK2100000054.

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https://doi.org/10.1016/j.jpowsour.2026.239941
Received 16 January 2026; Received in revised form 6 March 2026; Accepted 17 March 2026
Available online 25 March 2026
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defect engineering [8], continue to expand the material palette for energy conversion and storage applications. While these material innovations establish the foundation for high-performance fuel cells, accurate prognostics tools are equally essential to maximize the operational lifespan of PEMFC systems under realistic service conditions.

According to existing research, prognostics of PEMFC degradation can be broadly categorized into model-driven, data-driven, and hybrid approaches [9,10]. Model-driven methods rely on mechanistic understanding of physicochemical processes within PEMFC, describing aging behavior through empirical, semi-empirical, or degradation mechanism models [11]. Zhao et al. [12] developed an empirical decay model quantifying contributions of hydrogen permeation, membrane conductivity loss, and electrochemical active area reduction under dynamic cycling. Ou et al. [13] proposed a semi-empirical approach accounting for performance recovery and key parameter decay, validated under transient conditions. Zhou et al. [14] combined multi-physics aging models with particle filters for voltage degradation extrapolation. While offering low data dependency and strong generalization, model-driven approaches face challenges in capturing the full complexity of degradation mechanisms [15].

Data-driven methods construct prediction models using machine learning without requiring explicit degradation mechanisms, becoming increasingly prevalent [16]. Deng et al. [17] developed genetic algorithm-optimized stacked echo state networks for high-precision degradation prediction. Song et al. [18] integrated discrete Fréchet distance with particle swarm optimization-based extreme learning machine for state-of-health estimation. YAKUT et al. [19] demonstrated that single-step long short-term memory network (LSTM) predictions outperformed multi-step approaches under both dynamic and steady-state conditions. Zhang et al. [20] combined CNN feature extraction with LSTM hyperparameter optimization, improving both short- and long-term remaining useful life prediction. Jia et al. [21] employed bidirectional LSTM with multi-head self-attention for global information modeling of high-dimensional monitoring data. Despite excellent predictive performance, pure data-driven methods remain highly dependent on training data quality and quantity, lacking physical interpretability and often producing physically implausible predictions when extrapolating beyond observed data ranges.

Hybrid approaches integrate model-driven and data-driven strengths to enhance accuracy and robustness. Sun et al. [22] combined extended Kalman filters with multi-output Gaussian process regression for long-term performance prediction. Ma et al. [23] developed hybrid methods incorporating extended Kalman filters with LSTM networks for simultaneous voltage and aging parameter prediction. Ko et al. [24] established physics-informed neural network (PINN) frameworks embedding governing equations for membrane and catalyst degradation mechanisms. The integration of physical constraints into neural network training has emerged as a particularly promising paradigm, enabling the fusion of observational data with electrochemical priors to enhance generalization and ensure physical consistency in predictions [25].

Despite the success of attention-based models in time series forecasting, their direct application to PEMFC degradation prediction faces inherent challenges. PEMFC systems are characterized by multivariate operational parameters — including current density, temperature, humidity, and gas flow rates — that exhibit complex, coupled influences on voltage decay. Conventional approaches typically process such multivariate sequences by either treating each variable independently, thereby ignoring critical inter-parameter interactions, or flattening all dimensions into unified representations, which obscures the distinct physical meanings of individual operating conditions. For PEMFC systems driven by coupled physicochemical processes, these processing strategies result in suboptimal feature extraction, particularly under dynamic loads where current–temperature–humidity interactions are time-varying and nonlinear.

This paper integrates Crossformer [26] with physics-informed learning to address these limitations. Through Dimension-Segment-Wise (DSW) embedding and Two-Stage Attention (TSA), Crossformer preserves structural separation between time steps and operational variables, enabling explicit modeling of both cross-time dependencies and cross-dimension dependencies. The router-based attention mechanism efficiently captures these parameter interactions without quadratic complexity, while the hierarchical encoder–decoder structure extracts multi-scale features for both rapid transient responses and long-term degradation trends. Coupled with electrochemical constraints via a three-stage training strategy, this hybrid framework achieves robust prediction with limited training data across diverse operating conditions. The main innovations and contributions of this paper are as follows:

(1) **A hybrid prediction framework integrating data-driven and physical models is proposed**. Leveraging the convolution-enhanced Crossformer architecture, the method effectively captures complex interdependencies among different dimensions and enhances the capability to extract multi-channel degradation features, thereby enabling more accurate characterization of the time-dependent behavior of voltage decay.
(2) **Physical knowledge is introduced as guidance to enhance the model's robustness under data-limited conditions**. The adopted hybrid modeling strategy integrates data-driven learning with physical constraints, thereby effectively improving prediction performance and generalization when training samples are limited.
(3) **The model maintains low prediction errors and satisfactory accuracy across all current profiles, confirming its suitability for degradation forecasting under complex operating scenarios**.

The structure of this paper is organized as follows: Section 2 delineates the experimental dataset and data preprocessing procedures. Section 3 provides a comprehensive description of the hybrid forecasting model. The results and corresponding discussions are presented in Section 4. Finally, Section 5 concludes the paper by summarizing the key findings.

2. Dataset and data processing

2.1. PEMFC aging test data set

This study utilizes three groups of PEMFC aging test datasets, comprising steady-state, quasi-dynamic, and dynamic current operating conditions.

The steady-state and quasi-dynamic condition datasets are obtained from the PEMFC aging dataset [27] published by the French Fuel Cell Laboratory through the IEEE PHM 2014 Data Challenge. The experimental test bench is designed for fuel cell stacks with power ratings up to 1 kW, and durability testing was conducted over 1000 h on two independent stacks. Each stack consists of five individual cells with an active area of 100 cm^2 per cell, featuring a diffusion layer configuration. The cells operate at a nominal current density of 0.70 $A\,cm^{-2}$, with a maximum current density of 1 $A\,cm^{-2}$. The first experimental group involved stack operation under a constant current of 70 A for 1140 h, designated as static condition FC1. The second group tested the stack under a 70 A baseline current superimposed with a triangular waveform current disturbance (amplitude: 7 A, frequency: 5 kHz) for 1020 h, recorded as quasi-dynamic condition FC2.

The dataset under dynamic current conditions is sourced from the open-source durability experimental dataset provided by the Clean Energy Automotive Engineering Center and the School of Automotive Studies at Tongji University [28]. The experiment adopted the European harmonized testing protocol as the durability testing standard for PEMFC [29], and conducted dynamic durability and polarization characterization tests on the Greenlight 20 test station. The complete durability experiment comprised 3076 fuel cell dynamic load cycles

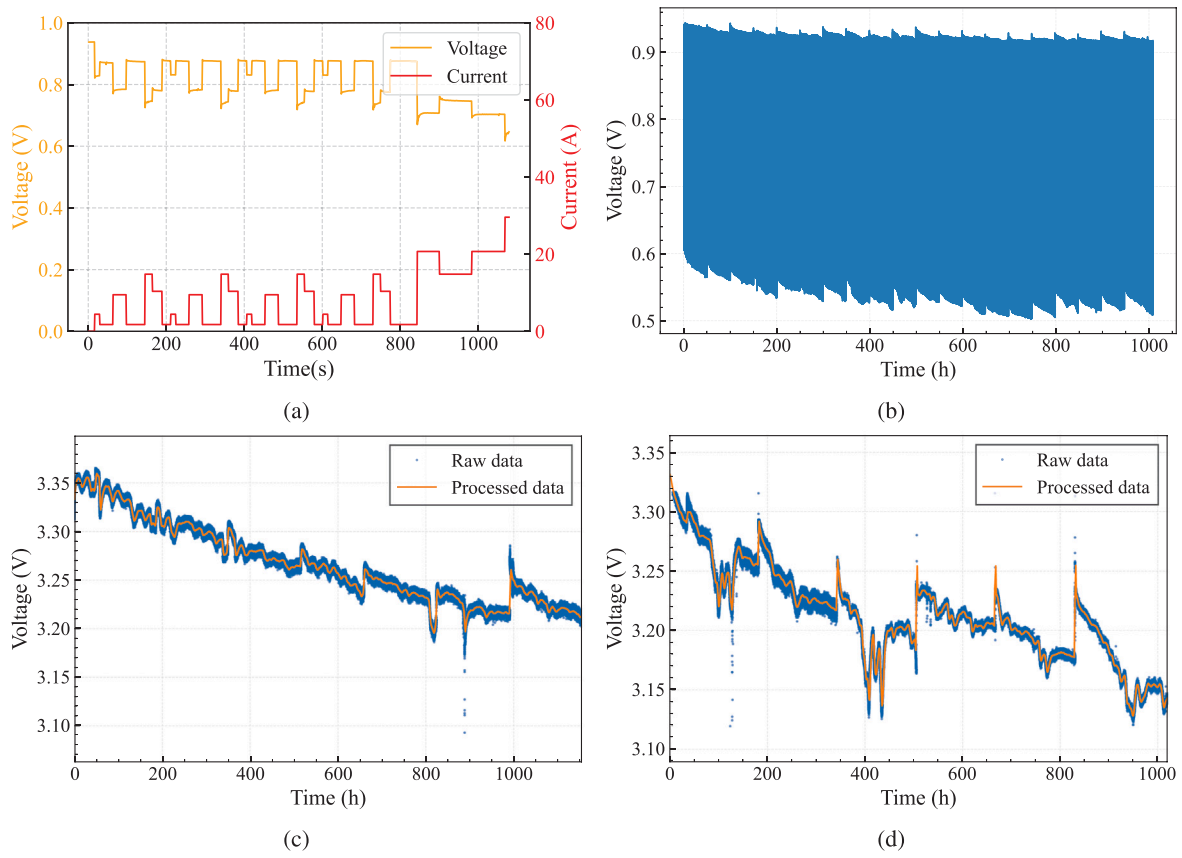


Fig. 1. Experimental data (a)Variation of current and voltage in an FC-DLC; (b)Voltage waveforms of FC3; (c) Raw, and processed voltages for FC1; (d) Raw, and processed voltages for FC2.

(FC-DLC), lasting approximately 1008 h. The sampling frequency for PEMFC durability testing was set at 1 Hz, with each FC-DLC lasting 1180 s and consisting of 35 testing phases. Voltage and current variations within a single FC-DLC are illustrated in Fig. 1(a). Polarization curve testing was performed after every 152 FC-DLCs (approximately 50 h). Additionally, a 12-hour shutdown period was implemented after every 100 h of operation to simulate the start-stop behavior of fuel cell vehicles, with this testing regime designated as dynamic condition FC3.

2.2. Data processing

The raw aging data of FC1 and FC2 contain substantial noise and spikes, which may interfere with the algorithm training process. Therefore, outlier processing and smoothing are required for the raw data. This paper employs the quartile method to handle outliers, defining an acceptable data range based on data quartiles, with any data points outside this range identified as outliers and subsequently removed. A moving average filter is applied for data smoothing, which essentially functions as a low-pass filter designed to eliminate high-frequency disturbances while preserving useful low-frequency trends in the time series. After filtering, given the large volume of original data, resampling is performed at 1-hour intervals to reconstruct samples, maintaining data variation trends while improving computational efficiency. Taking the voltage data of FC1 and FC2 as examples, the processed results are shown in Fig. 1(c) and Fig. 1(d).

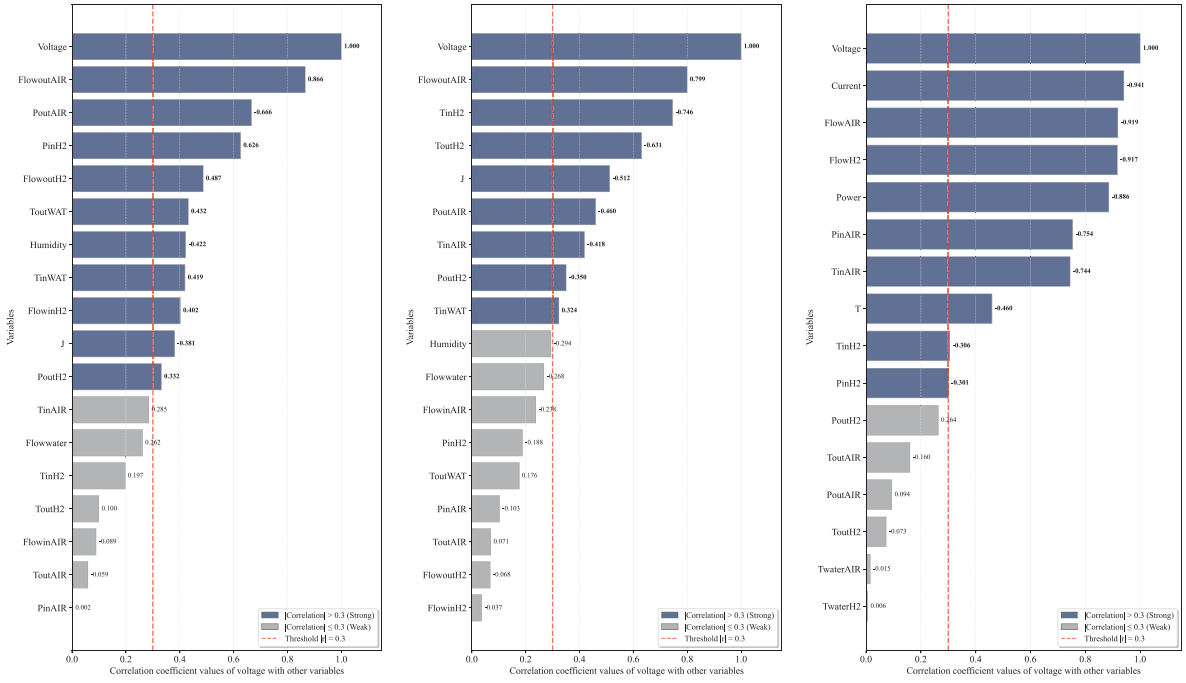
The original FC3 dataset was sampled at 1 Hz. To improve data processing efficiency, downsampling was performed on the raw data by selecting a single representative data point from each testing step to characterize the state of the PEMFC. Following this downsampling procedure, 35 data points were extracted from each FC-DLC cycle. This study adopts the downsampling strategy described in [28], which selects the fifth voltage value from the end of each testing step as the final representative value for that step in the FC-DLC. After downsampling, each FC-DLC cycle comprises 35 voltage points, with the resulting downsampled voltage data presented in Fig. 1(b).

2.3. Feature selection

Proton exchange membrane fuel cell data encompasses multiple parameters including temperature, pressure, flow rate, etc. However, excessive parameters in the prediction modeling process can lead to model redundancy and compromise prediction efficiency. Therefore, it is necessary to analyze and screen the experimentally obtained data, selecting the parameters most strongly correlated with voltage to balance computational speed and model accuracy. This study employs the Spearman rank correlation coefficient to evaluate interdependencies among data variables, with the corresponding equation expressed as follows:

$$\rho = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N\left(N^2-1\right)} \tag{1}$$

The strength of correlation between variables is positively correlated with the absolute values of their correlation coefficients. Variables demonstrating correlation coefficients below 0.3 with fuel cell voltage are considered weakly correlated parameters. These weakly correlated parameters are subsequently removed to reduce data dimensionality and enhance computational efficiency of the model. Fig. 2 shows the correlation coefficient values between voltage and other variables for FC1, FC2, and FC3 respectively, with the darker parts indicating the variables that have been screened.



(a) FC1 (b) FC2 (c) FC3

Fig. 2. Data filtering graph based on Spearman rank correlation coefficient value. (a) FC1; (b) FC2; (c) FC3.

3. Hybrid model prediction strategy

This section presents the implementation details and modular composition of the proposed hybrid prediction methodology, with the overall framework illustrated in Fig. 3.

3.1. Convolutional feature extractor

The convolutional layer is the core component of a CNN, which extracts features from the input feature map through the convolutional kernel. The nodes in each layer share the weights of the convolutional kernel, allowing the network to slide over the feature map while maintaining parameter invariance, thereby effectively identifying local fluctuation patterns in the data.

$$X_j^l = \sum_{i=1}^{M} X_i^{l-1} * K_{ij}^l + b_j^l, \quad j = 1, 2, \ldots, N \tag{2}$$

where M denotes the number of input feature maps, N represents the number of convolutional kernels, X_j^l is the jth output feature map at layer l, X_i^{l-1} indicates the ith input feature map from layer $l-1$, K_{ij}^l denotes the kernel parameters connecting the ith input feature to the jth output feature, and b_j^l is the bias term corresponding to the jth convolutional kernel.

In this study, to effectively capture local temporal patterns and dynamic features in the voltage time series of PEMFC, we designed a specialized one-dimensional convolutional feature extractor. The primary function of this module is to extract discriminative local features from multivariate time series data, thereby providing high-quality input representations for subsequent prediction models.

3.2. Crossformer

Crossformer is a Transformer-based model specifically designed for multivariate time series forecasting, with its core innovation lying in the explicit modeling of both cross-time and cross-dimension dependencies. The architecture primarily comprises three key components: the DSW embedding module, the TSA layer, and a Hierarchical Encoder–Decoder (HED) structure. The structural diagram is shown in Fig. 4.

Unlike conventional Transformer model that embed all dimension data at a single timestep into one unified vector — primarily capturing temporal dependencies while neglecting inter-dimensional correlations — Crossformer introduces a Dimension-Segment-Wise embedding approach to preserve both temporal and dimensional information. Specifically, this method partitions the time series of each dimension into segments of length L_{seg}, and transforms each segment into a representative vector through linear projection and positional encoding techniques:

$$\mathbf{h}_{i,d} = \mathbf{E}\mathbf{x}_{i,d}^{(s)} + \mathbf{E}_{i,d}^{(pos)} \tag{3}$$

where $\mathbf{x}_{i,d}^{(s)} \in \mathbb{R}^{Lseg}$ represents the ith segment of the dth dimension, $\mathbf{E} \in \mathbb{R}^{d_{model} \times L_{seg}}$ denotes the learnable projection matrix, $\mathbf{E}_{i,d}^{(pos)} \in \mathbb{R}^{d_{model}}$ denotes the learnable position embedding for position (i, d), and $\mathbf{h}i, d \in \mathbb{R}^{d_{model}}$ is the embedded vector for the segment.

This produces a 2D vector array

$$\mathbf{H} = \{\mathbf{h}i, d | 1 \le i \le L, 1 \le d \le D\} \in \mathbb{R}^{L \times D \times d_{model}} \tag{4}$$

where $L = T/L_{seg}$ is the number of segments per dimension, D is the number of dimensions (variables) in the multivariate time series, and d_{model} is the hidden dimension size.

To efficiently capture both cross-time and cross-dimension dependencies, Crossformer designs a Two-Stage Attention layer.

Cross-Time Stage: Multi-head self-attention is applied separately to each dimension:

$$\hat{\mathbf{Z}}_{i,d}^{time} = \text{LayerNorm}\left(\mathbf{Z}_{i,d} + \text{MSA}^{time}(\mathbf{Z}_{i,d}, \mathbf{Z}_{i,d}, \mathbf{Z}_{i,d})\right) \tag{5}$$

In this formulation: $\mathbf{Z}_{i,d} \in \mathbb{R}^{L \times d_{model}}$ encompasses all segment vectors of dimension d at time step i, MSA^{time} denotes the multi-head self-attention operation specialized for temporal analysis, and $\hat{\mathbf{Z}}_{i,d}^{time}$ represents the normalized output following attention computation. This stage operates with computational complexity $O(DL^2)$, where parameter sharing across dimensions significantly optimizes computational efficiency.

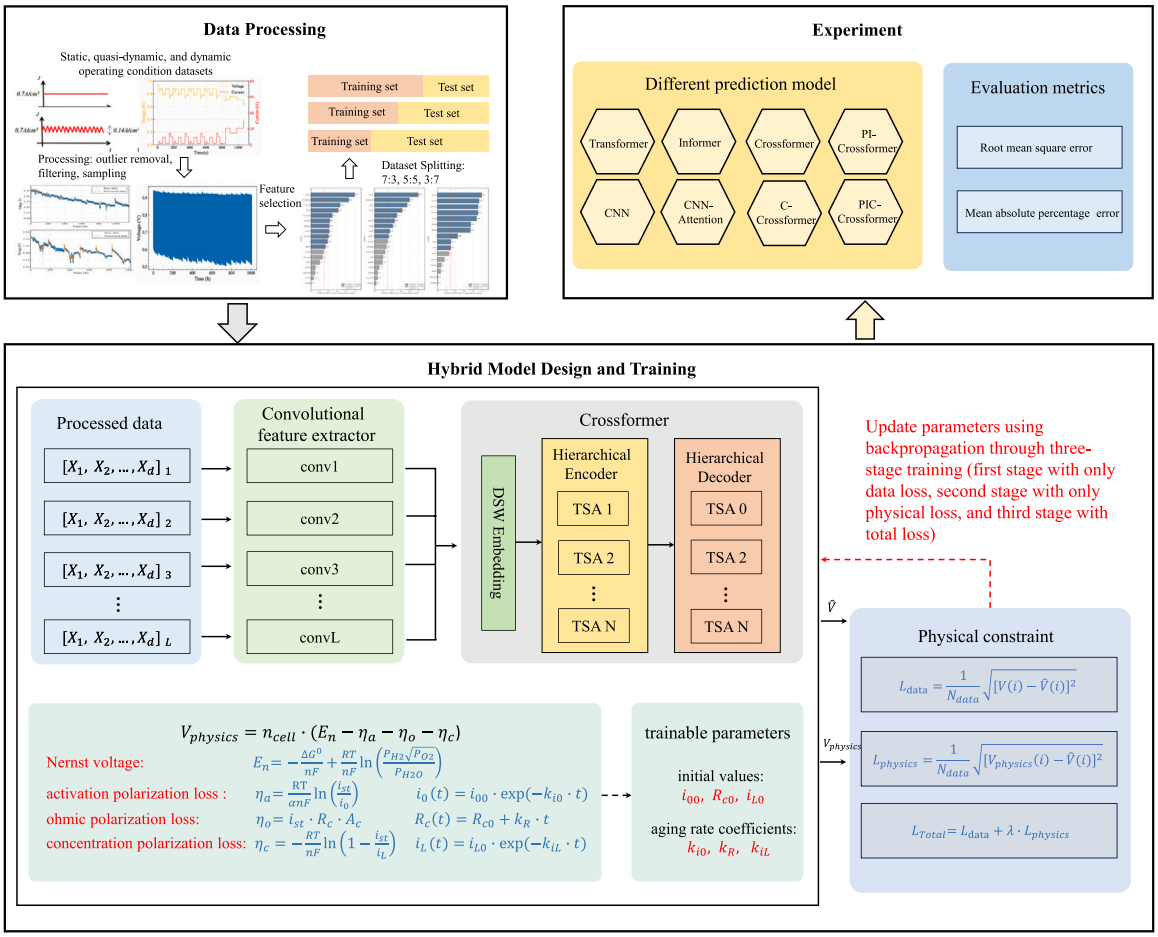


Fig. 3. Overall framework diagram of hybrid prediction method.

Cross-Dimension Stage: To avoid the direct computation of attention among all dimensions, which would incur $O(D^2)$ complexity, a router mechanism is introduced. Specifically, for each time step, c learnable router vectors are employed to first aggregate information from all dimensions and then redistribute it:

$$\mathbf{B}_{i,:} = MSA_1^{dim}(\mathbf{R}_{i,:}, \mathbf{Z}_{i,:}^{time}, \mathbf{Z}_{i,:}^{time}) \tag{6}$$

$$\overline{\mathbf{Z}}_{i,:}^{dim} = MSA_2^{dim}(\mathbf{Z}_{i,:}^{time}, \mathbf{B}_{i,:}, \mathbf{B}_{i,:}) \tag{7}$$

Here, the router vector array $\mathbf{R} \in \mathbb{R}^{L\times c\times d_{model}}$ functions as an information bottleneck, $\mathbf{B}_{i,:} \in \mathbb{R}^{c\times d_{model}}$ encapsulates aggregated information from all dimensions, and $\overline{\mathbf{Z}}^{dim} i,: \in \mathbb{R}^{D\times d_{model}}$ denotes the refined output after cross-dimensional attention processing. This architectural innovation reduces the stage complexity to $O(cDL)$, maintaining the overall Two-Stage Attention layer complexity at $O(DL^2)$.

To further leverage multi-scale information, Crossformer constructs a Hierarchical Encoder–Decoder structure.

Encoder: Each layer merges adjacent temporal vectors to form coarser-grained representations:

$$\hat{\mathbf{Z}}_{i,d}^{enc,l} = \mathbf{M}[\mathbf{Z}_{2i-1,d}^{enc,l-1} \cdot \mathbf{Z}_{2i,d}^{enc,l-1}] \tag{8}$$

where $\mathbf{M} \in \mathbb{R}^{d_{model}\times 2d_{model}}$ denotes the learnable matrix for segment merging, $\mathbf{Z}^{enc,l-1}$ corresponds to the output features from the previous encoder layer, and $\hat{\mathbf{Z}}^{enc,l}$ indicates the merged intermediate representation at layer l. The TSA layer then captures scale-specific dependencies: $\mathbf{Z}^{enc,l} = \mathrm{TSA}(\hat{\mathbf{Z}}^{enc,l})$.

Decoder: Each layer receives the corresponding scale output from the encoder and generates predictions through TSA and cross-attention:

$$\overline{\mathbf{Z}}_{i,d}^{dec,l} = \mathrm{MSA}(\hat{\mathbf{Z}}_{i,d}^{dec,l}, \mathbf{Z}_{i,d}^{enc,l}, \mathbf{Z}_{i,d}^{enc,l}) \tag{9}$$

where $\hat{\mathbf{Z}}^{dec,l}$ denotes the output features from the TSA module in the lth decoder layer, $\mathbf{Z}^{enc,l}$ represents the encoder output features at the same scale l, and $\overline{\mathbf{Z}}^{dec,l}$ corresponds to the output representation after encoder–decoder cross-attention. The final prediction is obtained through linear projection and hierarchical aggregation of all layers:

$$\mathbf{x}_{T+1:T+\tau}^{pred} = \sum_{l=0}^{N} \mathbf{x}_{T+1:T+\tau}^{pred,l} \tag{10}$$

$$\mathbf{x}_{T+1:T+\tau}^{pred,l} = \mathbf{W}^l \mathbf{Z}_{i,d}^{dec,l} \mid 1 \le i \le \tau/L_{seg}, 1 \le d \le D \tag{11}$$

Eq. (11) indicates the prediction from the lth decoder layer, $\mathbf{W}^l \in \mathbb{R}^{L_{seg}\times d_{model}}$ is a scale-specific learnable projection matrix, N represents the total number of decoder layers, and τ denotes the prediction horizon. This hierarchical architecture effectively captures both short-term details and long-term trends through multi-scale feature fusion, significantly enhancing the accuracy of long-term time series forecasting.

3.3. PEMFC degradation model

To accurately characterize the performance degradation process of proton exchange membrane fuel cells, an aging model can be established based on stack voltage decay behavior. This model comprehensively incorporates key aging mechanisms including catalyst activity decay, proton exchange membrane impedance increase, and gas diffusion layer performance degradation. The output voltage of a fuel cell

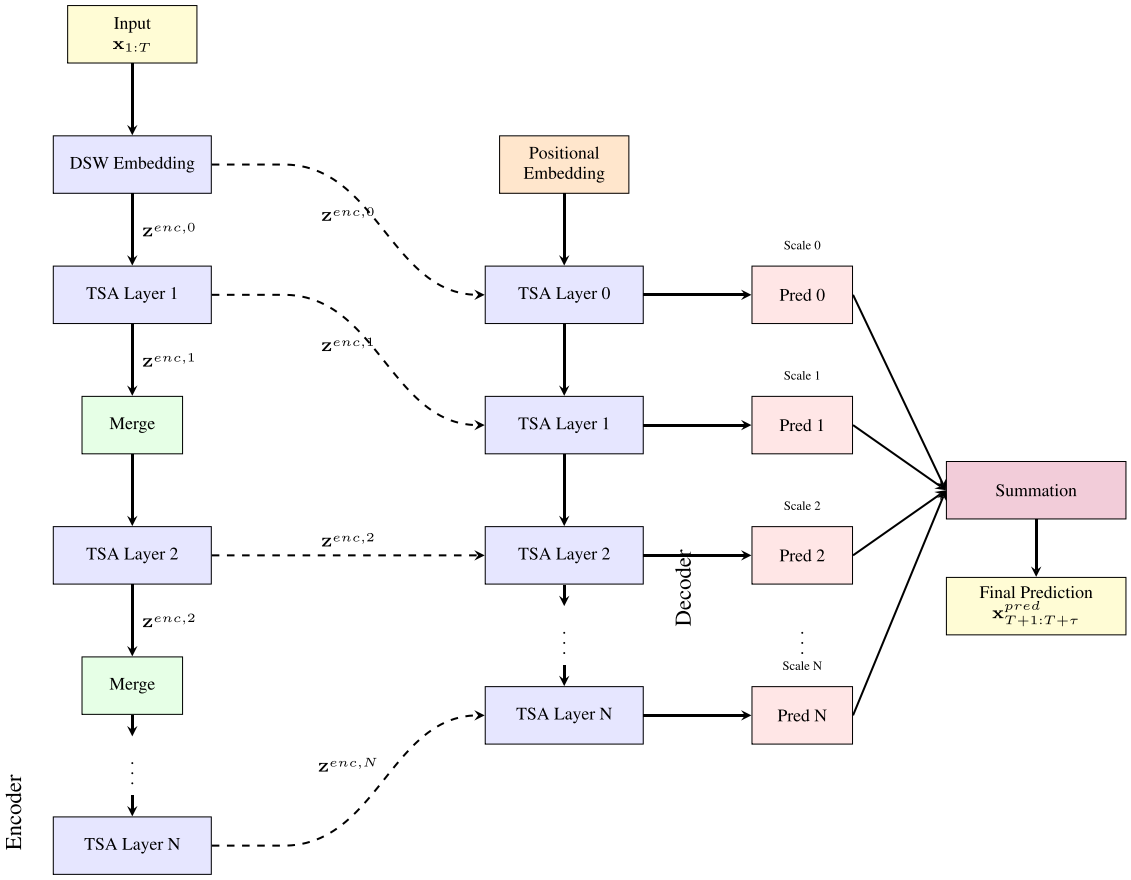


Fig. 4. Schematic diagram of the Crossformer structure.

stack can be represented as the Nernst voltage minus various voltage losses [30]:

$$V_{\text{stack}} = n_{\text{cell}} \cdot (E_{\text{n}} - \eta_{\text{a}} - \eta_{\text{o}} - \eta_{\text{c}}) \tag{12}$$

where n_{cell} denotes the number of single cells, E_{n} represents the Nernst voltage (V), while η_{a}, η_{o}, and η_{c} correspond to the activation polarization loss, ohmic polarization loss, and concentration polarization loss, respectively.

The Nernst voltage is calculated as follows:

$$E_{\text{n}} = -\frac{\Delta G^0}{nF} + \frac{RT}{nF} \ln\left(\frac{P_{\text{H}_2}\sqrt{P_{\text{O}_2}}}{P_{\text{H}_2\text{O}}}\right) \tag{13}$$

where ΔG^0 denotes the Gibbs free energy change per mole of reaction (J/mol), n represents the number of electrons transferred per mole of reaction (mol^{-1}), F = 96485 C/mol is the Faraday constant, R = 8.314 J/(mol·K) is the universal gas constant, T indicates the stack temperature (K), and P_{H2}, P_{O2}, and $P_{\text{H}_2\text{O}}$ correspond to the partial pressures (bar) of hydrogen, oxygen, and water vapor, respectively.

The three components of voltage losses are expressed as functions of operating conditions and aging parameters [31]:

$$\begin{cases} \eta_{\text{a}} = \dfrac{RT}{\alpha nF} \ln\left(\dfrac{i_{\text{st}}}{i_0}\right) \\ \eta_{\text{o}} = i_{\text{st}} \cdot R_c \cdot A_c \\ \eta_{\text{c}} = -\dfrac{RT}{nF} \ln\left(1 - \dfrac{i_{\text{st}}}{i_L}\right) \end{cases} \tag{14}$$

where i_{st} is the current density (A/cm^2), A_c is the active area (cm^2), α is the charge transfer coefficient, and i_0, R_c, and i_L are the three key aging-related parameters: exchange current density (A/cm^2), cell resistivity (Ω), and limiting current density (A/cm^2).

Considering the nonlinear characteristics of the aging process, **the three key aging parameters vary with time according to the following model**:

$$\begin{cases} i_0(t) = i_{00} \cdot \exp(-k_{i0} \cdot t) \\ R_{\text{c}}(t) = R_{c0} + k_R \cdot t \\ i_L(t) = i_{L0} \cdot \exp(-k_{iL} \cdot t) \end{cases} \tag{15}$$

where i_{00}, R_{c0}, and i_{L0} are the initial values of the parameters, k_{i0}, k_R, and k_{iL} are the corresponding aging rate coefficients, and t is the operating time (h). This modeling approach effectively captures the nonlinear decay trends of aging parameters over time.

3.4. PINN

PINN is a model architecture designed to embed physical laws (typically described by partial differential equations) as prior knowledge into the neural network training process. By incorporating governing equations, initial conditions, boundary conditions, and potential experimental observations into the loss function, PINN can learn and approximate the dynamic principles of physical systems in an unsupervised or weakly supervised manner [32]. Unlike conventional deep learning models that rely solely on data-driven approaches, PINN introduces physical constraints into the loss function by constructing a physical residual term. This is achieved by enforcing the neural network predictions to satisfy the governing equations, specifically through evaluating equation residuals at a set of collocation points within the computational domain, which subsequently participates in backpropagation as part of the optimization objective. Consequently, PINN simultaneously considers two types of losses during training: the first is data loss, measuring the discrepancy between predictions and actual measurements; the second is physical loss, characterizing

the deviation of predicted solutions from the governing equations. This integrated training paradigm combining data-driven learning with physical mechanisms enables PINN to effectively fuse observational information with physical priors, thereby more accurately approximating the latent functions reflecting degradation trends in aging test data while enhancing model generalization and physical consistency.

Building upon the theoretical foundation of PINN, this work integrates the PEMFC voltage decay model described in Section 3.3 with the Crossformer architecture to predict PEMFC output voltage. As shown in Fig. 3, with the loss functions comprising data loss (L_{data}) and physical loss (L_{physics}), whose summation constitutes the total loss (L_{Total}):

$$\begin{cases} L_{\text{data}} = \dfrac{1}{N_{\text{data}}}\sqrt{[V(i)-\hat{V}(i)]^2} \\ L_{\text{physics}} = \dfrac{1}{N_{\text{data}}}\sqrt{[V_{\text{physics}}(i)-\hat{V}(i)]^2} \\ L_{\text{Total}} = L_{\text{data}} + \lambda \cdot L_{\text{physics}} \end{cases} \tag{16}$$

where V denotes the actual measured voltage, $\hat{V}$ represents the predicted voltage from the model, V_{physics} indicates the physics-constrained voltage value computed from Eq. (12), and λ is the weighting coefficient that balances the relative importance between data loss and physical loss in the total loss function.

3.5. Voltage prediction strategy

This study utilizes preprocessed datasets divided in a 7:3 ratio for training and testing, covering three distinct current load conditions (FC1, FC2, and FC3). As shown in Fig. 3, the proposed physics-informed convolutional-enhanced Crossformer (PIC-Crossformer) model first processes multidimensional operational time-series data of PEMFC through convolutional feature extractors for local feature extraction. The Crossformer model then employs cross-attention mechanisms across both temporal and feature dimensions to capture long-term dependencies and cross-dimensional correlations. A physics-constrained layer is subsequently incorporated to embed PEMFC electrochemical mechanisms into the neural network architecture. To enhance training efficiency and mitigate the risk of converging to local optima, we employ a three-stage progressive training strategy. Direct end-to-end optimization of the compound loss function L_{Total} often suffers from slow convergence and instability, as the data-driven and physics-based components may compete during gradient updates, leading to suboptimal solutions. Our strategy addresses this by decomposing the optimization process: the first stage initializes model parameters using pure data loss to rapidly capture degradation trends; the second stage performs unsupervised refinement via physical loss to regularize aging parameter evolution; the third stage conducts joint fine-tuning for synergistic optimization. Battery voltage prediction is achieved through a sliding window approach (window size = 24) using a multiple-input–single-output architecture for one-step short-term forecasting of performance degradation across all three current conditions in the FC datasets.

4. Results and discussions

4.1. Evaluation indicators

Appropriate quantitative metrics are essential for objectively evaluating predictive performance. This hybrid prediction model employs RMSE and mean MAPE as evaluation criteria, where values closer to zero indicate higher prediction accuracy. The specific definitions are as follows:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2} \tag{17}$$

$$MAPE = \frac{100\%}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{18}$$

where y_i, $\hat{y}_i$ represent the real and predicted voltage values, respectively; N denotes the number of real voltages.

4.2. Comparative experiments

We systematically compared PIC-Crossformer with two types of baseline models, CNN and Transformer, along with two advanced architectures, Informer [33] and CNN-Attention [34]. Fig. 5 illustrates the predictive results of different models under steady-state (FC1), quasi-dynamic (FC2), and fully dynamic (FC3) current profiles, with training-set ratios of 70%, 50%, and 30%, while the corresponding quantitative metrics are consolidated in Table 1.

The experimental results show that under steady-state, quasi-dynamic, and fully dynamic current regimes, PIC-Crossformer achieves **RMSEs of 0.00191, 0.00324, and 0.00305** at 70% training split, with corresponding **MAPEs of 0.0339%, 0.0406%, and 0.323%**, respectively. Except for the condition where the training set accounts for 30% under FC2 operating conditions, where the CNN-Attention [34] model demonstrates slightly superior prediction accuracy compared to the PIC-Crossformer model, the PIC-Crossformer consistently achieves the lowest RMSE and MAPE across all other experimental scenarios. These comprehensive results substantiate the superior predictive performance of the PIC-Crossformer hybrid model in accurately forecasting PEMFC voltage under diverse operating conditions, while simultaneously validating its exceptional generalization capabilities and practical engineering application potential.

To evaluate the model's capability for longer prediction horizons, we extend the forecasting task from one-step to multi-step prediction. Table 2 presents the predictive performance of different models at 1-step, 5-step, and 10-step horizons under 70% training data ratio. As expected, prediction errors accumulate with increasing forecast horizon due to the autoregressive nature of sequential forecasting. For PIC-Crossformer, RMSE increases from 0.00191 V (1-step) to 0.00585 V (5-step) and 0.00755 V (10-step) on FC1, with corresponding MAPE rising from 0.0339% to 0.101% and 0.142%. Similar degradation patterns are observed on FC2 and FC3. Notably, PIC-Crossformer maintains relative performance advantages across all horizons compared to baseline models. On FC3, where dynamic loads introduce highest variability, PIC-Crossformer achieves 10-step RMSE of 0.0273 V versus 0.0851 V for Transformer and 0.0371 V for CNN-Attention, demonstrating the robustness conferred by physics-informed constraints in mitigating error propagation. The hierarchical multi-scale structure of Crossformer contributes to this stability by capturing both short-term transients and long-term trends essential for medium-horizon forecasting.

4.3. Ablation experiments

To systematically investigate the individual and synergistic impacts of convolutional modules and physical constraint mechanisms on prediction accuracy, we designed and conducted a series of controlled ablation experiments. Specifically, we constructed and compared four model variants: PIC-Crossformer, standard baseline Crossformer model, Convolutional-enhanced Crossformer model (C-Crossformer) with only architectural modifications, and Physics-Informed Crossformer model (PI-Crossformer) embedding solely physical constraints. Through quantitative analysis of performance disparities under identical experimental conditions, we effectively decomposed the relative contributions of convolutional feature extraction and physics-informed regularization to the model's predictive capability. Additionally, we evaluated model performance with limited training data by using training set proportions of 70%, 50%, and 30%. The prediction results under FC1, FC2, and FC3 current conditions are shown in Fig. 6, with detailed performance metrics provided in Table 3.

Experimental results demonstrate a strict hierarchical relationship in prediction accuracy across three operating conditions: the PIC-Crossformer model consistently outperforms the PI-Crossformer model, which significantly surpasses the C–Crossformer model, while the C–Crossformer model stably exceeds the baseline Crossformer model. This monotonically increasing performance gradient comprehensively

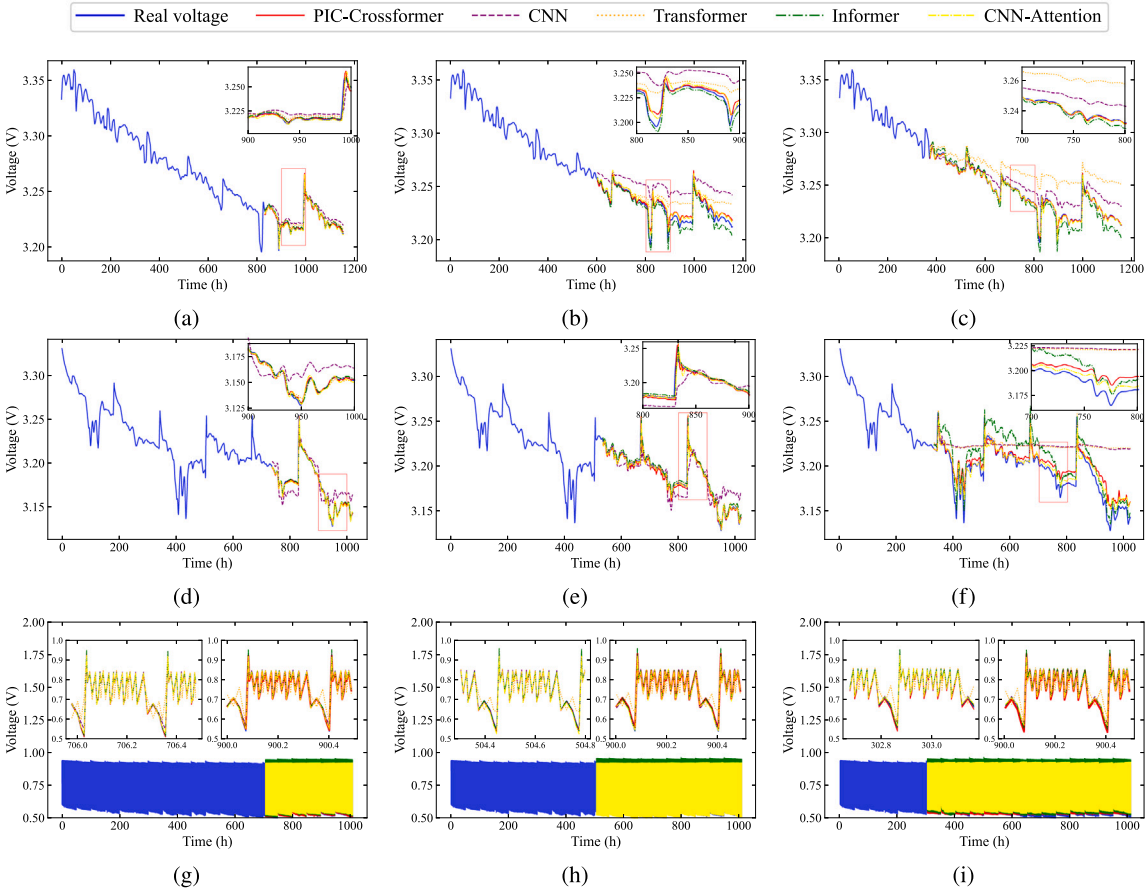


Fig. 5. Prediction results of different models under FC1 (a–c), FC2 (d–f) and FC3 (g–i) with training-set ratios of 70%, 50% and 30%.

Table 1
Prediction performance metrics of different models on FC1, FC2, and FC3 under various training set proportions.

Dataset	Model	Training set accounts for 70%		Training set accounts for 50%		Training set accounts for 30%	
		RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)
FC1	Transformer	0.00299	0.0692	0.0118	0.289	0.0239	0.660
	CNN	0.00609	0.152	0.0208	0.596	0.0109	0.271
	Informer [33]	0.00219	0.0479	0.00483	0.122	0.00573	0.140
	CNN-Attention [34]	0.00222	0.0360	0.00435	0.111	0.00254	0.0412
	PIC-Crossformer	0.00191	0.0339	0.00402	0.0909	0.00218	0.0325
FC2	Transformer	0.00397	0.0678	0.00441	0.0910	0.0380	0.952
	CNN	0.0131	0.356	0.0138	0.318	0.0381	0.959
	Informer [33]	0.00350	0.0443	0.00426	0.0986	0.0163	0.472
	CNN-Attention [34]	0.00362	0.0520	0.00378	0.0506	0.00982	0.224
	PIC-Crossformer	0.00324	0.0406	0.00361	0.0632	0.0133	0.331
FC3	Transformer	0.0432	4.36	0.0608	5.66	0.0649	5.64
	CNN	0.00930	0.868	0.0115	1.32	0.0210	2.41
	Informer [33]	0.00655	0.466	0.00712	0.547	0.0116	1.34
	CNN-Attention [34]	0.00405	0.418	0.00613	0.656	0.0110	1.19
	PIC-Crossformer	0.00305	0.323	0.00543	0.583	0.0102	1.15

validates the synergistic integration of convolutional feature extractors and physical constraint mechanisms for enhancing model predictive capability.

To further elucidate the interaction between convolutional modules and physics constraints, we analyze their compensatory behavior under varying data availability. At 70% training data ratio, both components contribute additively: the convolutional module achieves 16.8% RMSE reduction over baseline Crossformer by extracting local temporal patterns, while physics constraints alone yield 28.4% improvement through solution space regularization. Their combined effect in PIC-Crossformer produces supra-additive gains, indicating synergistic interaction where convolutional features enhance physical parameter identification while constraints stabilize predictions against overfitting. When training data is reduced to 30%, asymmetric compensation emerges. Removing convolutional modules (PI-Crossformer versus PIC-Crossformer) causes 52% performance degradation on FC1, whereas removing physics constraints (C-Crossformer versus PIC-Crossformer) leads to 78% degradation. This asymmetry demonstrates that physics

Table 2
Predictive performance metrics of different models on FC1, FC2, and FC3 at different prediction steps.

Dataset	Model	1-step		5-step		10-step	
		RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)
FC1	Transformer	0.00299	0.0692	0.00640	0.0138	0.00845	0.195
	CNN	0.00609	0.152	0.0114	0.295	0.0149	0.396
	Informer [33]	0.00219	0.0479	0.00624	0.132	0.00820	0.168
	CNN-Attention [34]	0.00222	0.0360	0.00668	0.113	0.00913	0.187
	PIC-Crossformer	0.00191	0.0339	0.00585	0.101	0.00755	0.142
FC2	Transformer	0.00397	0.0678	0.0120	0.271	0.0167	0.465
	CNN	0.0131	0.356	0.0190	0.457	0.0222	0.569
	Informer [33]	0.00350	0.0443	0.0107	0.251	0.0141	0.366
	CNN-Attention [34]	0.00362	0.0520	0.0111	0.255	0.0149	0.332
	PIC-Crossformer	0.00324	0.0406	0.00994	0.210	0.0134	0.344
FC3	Transformer	0.0432	4.36	0.0783	8.45	0.0851	9.58
	CNN	0.00930	0.868	0.0190	2.03	0.0371	3.71
	Informer [33]	0.00655	0.466	0.0753	8.39	0.0817	9.33
	CNN-Attention [34]	0.00405	0.418	0.0171	1.78	0.0336	3.61
	PIC-Crossformer	0.00305	0.323	0.0164	1.65	0.0273	2.50

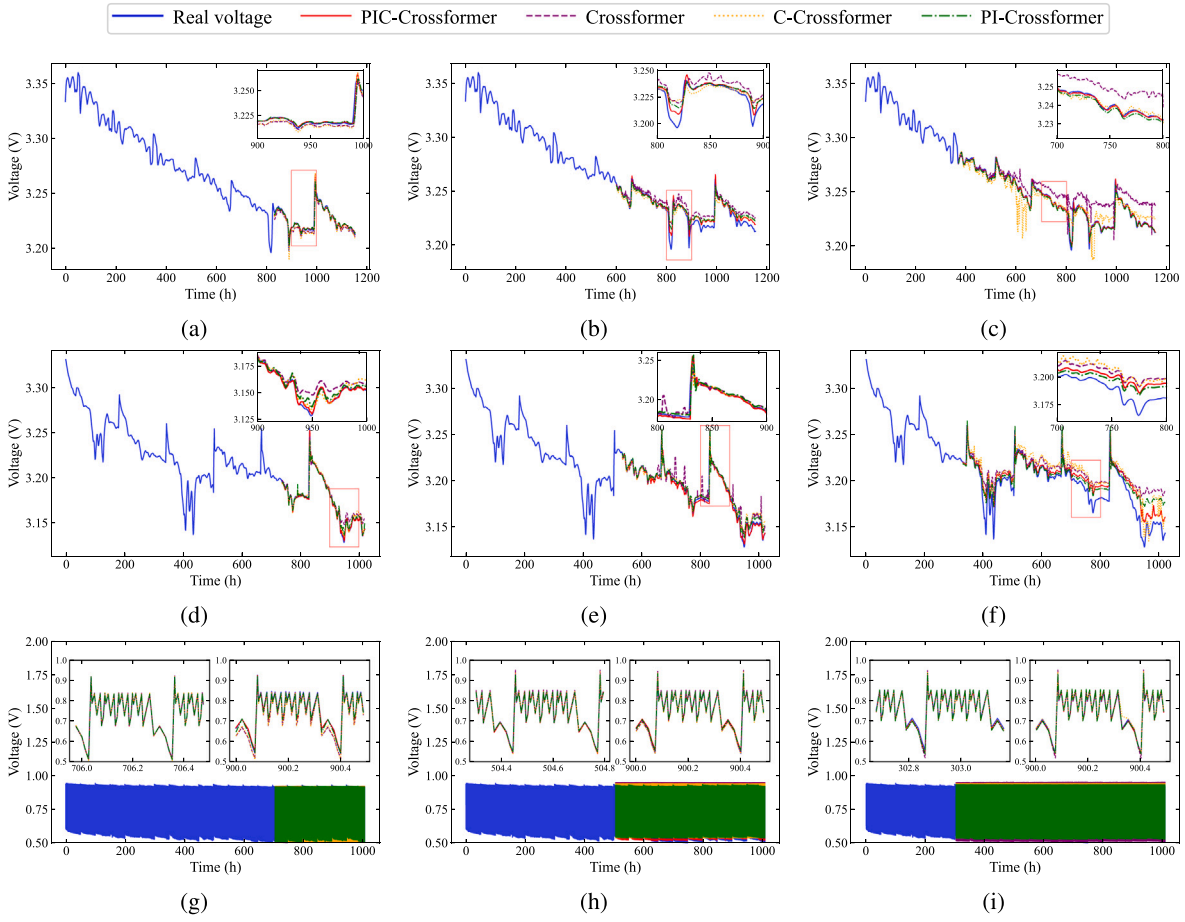


Fig. 6. Results of ablation experiments under FC1 (a–c), FC2 (d–f) and FC3 (g–i) with training-set ratios of 70%, 50% and 30%.

constraints provide stronger regularization under data scarcity, compensating for convolutional overfitting susceptibility. The performance gap between PIC-Crossformer and C–Crossformer increases from a ratio of 1.7 to 3.3 as the training data proportion decreases from 70% to 30%, whereas the gap between PIC-Crossformer and PI-Crossformer remains consistent. This demonstrates that physical constraints act as the primary compensatory mechanism when data-driven feature extraction becomes unreliable, thereby validating the integrated design's effectiveness for practical deployment with limited training samples.

Specifically, under FC1 conditions, the PIC-Crossformer model maintains minimal degradation in RMSE and MAPE metrics when the training set proportion decreases to 50% and 30%, compared to the 70% baseline. Similarly, under FC2 and FC3 conditions, the prediction accuracy of the PIC-Crossformer model at 50% training ratio shows negligible difference from the 70% scenario; although performance exhibits moderate decline at 30% proportion, it remains at a high precision level. These collective findings indicate that the PIC-Crossformer model preserves robust prediction performance under scarce training data

Table 3
Ablation experiment prediction performance metrics for FC1, FC2, and FC3 under different training set proportions.

Dataset	Model	Training set accounts for 70%		Training set accounts for 50%		Training set accounts for 30%	
		RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)	RMSE (V)	MAPE (%)
FC1	Crossformer	0.00334	0.0784	0.00763	0.194	0.0131	0.320
	C-Crossformer	0.00278	0.0681	0.00590	0.133	0.00880	0.182
	PI-Crossformer	0.00239	0.0366	0.00536	0.120	0.00260	0.0513
	PIC-Crossformer	0.00191	0.0339	0.00402	0.0909	0.00218	0.0325
FC2	Crossformer	0.00695	0.139	0.00734	0.1465	0.0204	0.493
	C-Crossformer	0.00576	0.110	0.00580	0.119	0.0155	0.425
	PI-Crossformer	0.00399	0.0665	0.00470	0.0927	0.0144	0.303
	PIC-Crossformer	0.00324	0.0406	0.00361	0.0632	0.0133	0.331
FC3	Crossformer	0.0112	1.15	0.0151	1.76	0.0180	2.11
	C-Crossformer	0.0109	1.11	0.0117	1.29	0.0174	2.03
	PI-Crossformer	0.00417	0.446	0.00700	0.673	0.0143	1.68
	PIC-Crossformer	0.00305	0.323	0.00543	0.583	0.0102	1.15

conditions, validating its exceptional sample efficiency and few-shot learning capability.

Electrochemical models impose strong inductive biases by constraining predictions within physically plausible degradation trajectory spaces. In contrast to purely data-driven models that operate in unconstrained hypothesis spaces, hybrid models intrinsically capture the fundamental physical processes including catalyst activity decay, gas diffusion layer performance degradation, and membrane resistance increment, thereby effectively preventing predictions that violate fundamental electrochemical principles. This physical constraint mechanism becomes particularly crucial under scarce training data scenarios, where it significantly reduces the effective complexity of the learning problem through information-theoretic regularization, consequently enhancing both model generalization capability and prediction reliability.

5. Conclusions

This paper proposes a hybrid model (PIC-Crossformer) for PEMFC aging prediction . By coupling the Crossformer architecture with explicit physics-based constraints, the model achieves a synergistic enhancement between data-driven learning capacity and electrochemical prior knowledge. The framework begins with preprocessing health monitoring data, followed by feeding historically measured voltage data and operational parameter sequences into the model using a sliding window approach. The hybrid model employs a convolutional feature extractor to capture local patterns in the data, while leveraging the cross-attention mechanism of Crossformer to simultaneously model long-term dependencies and cross-dimensional relationships in both temporal and feature domains. Comprehensive comparisons with other deep learning prediction models validate the effectiveness of the proposed approach. Furthermore, ablation studies quantitatively demonstrate the contribution of convolutional feature extractors and physical equation embedding to prediction accuracy improvement. **Experimental results confirm that the model achieves satisfactory prediction performance with limited training data under different current conditions**. Future work will focus on refining the fusion architecture to enhance predictive capability and extending the model to long-term forecasting scenarios.

CRediT authorship contribution statement

Lin Wei: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Shangrui Gao:** Writing – review & editing, Data curation. **Zonghai Chen:** Writing – review & editing, Investigation. **Yujie Wang:** Supervision, Resources, Funding acquisition. **Zhendong Sun:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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